

* Rank 0 tensor.

$$[G]$$

$$G = \mathbb{G}_0^{(1)}$$

* Rank 1 tensor.

$$[G_x \quad G_y \quad G_z]$$

$$G_x = \mathbb{G}_x^{(1)}$$

$$G_y = \mathbb{G}_y^{(1)}$$

$$G_z = \mathbb{G}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & G_{xy} & G_{xz} \\ G_{xy} & G_{yy} & G_{yz} \\ G_{xz} & G_{yz} & G_{zz} \end{bmatrix}$$

$$G_{xx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$

$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$

$$G_{xz} = \mathbb{G}_{xz}^{(1)}$$

$$G_{yy} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$

$$G_{yz} = \mathbb{G}_{yz}^{(1)}$$

$$G_{zz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & G_{xz} \\ -G_{xy} & 0 & G_{yz} \\ -G_{xz} & -G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}_z^{(1)}$$

$$G_{xz} = -\mathbb{Q}_y^{(1)}$$

$$G_{yz} = \mathbb{Q}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} G_{xxx} & G_{xxy} & G_{xxz} \\ G_{yyx} & G_{yyy} & G_{yyz} \\ G_{zzx} & G_{zzy} & G_{zzz} \\ G_{yzx} & G_{yyz} & G_{zzz} \\ G_{xxz} & G_{yzx} & G_{zzz} \\ G_{xxy} & G_{yyx} & G_{yzx} \end{bmatrix}$$

$$\begin{aligned}
G_{xxx} &= 3\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)} \\
G_{xxy} &= \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)} \\
G_{xxz} &= \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\beta(1)} \\
G_{yyx} &= \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\beta(1)} \\
G_{yyy} &= 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} \\
G_{yyz} &= \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\beta(1)} \\
G_{zzx} &= \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} \\
G_{zzy} &= \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)} \\
G_{zzz} &= 3\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)} \\
G_{yzx} &= \mathbb{G}_{xyz}^{(1)}
\end{aligned}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & G_{xxz} \\ G_{yyx} & 0 & G_{yyz} \\ G_{zzx} & G_{zz y} & 0 \\ G_{yzx} & -\frac{G_{yyz}}{2} & -\frac{G_{zz y}}{2} \\ -\frac{G_{xxz}}{2} & G_{zxy} & -\frac{G_{zzx}}{2} \\ -\frac{G_{xxy}}{2} & -\frac{G_{yyx}}{2} & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$\begin{aligned}
G_{xxy} &= 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)} \\
G_{xxz} &= 2\mathbb{G}_z^{(2)} + 2\mathbb{Q}_{xy}^{(1)} \\
G_{yyx} &= 2\mathbb{G}_x^{(2)} + 2\mathbb{Q}_{yz}^{(1)} \\
G_{yyz} &= 2\mathbb{G}_z^{(2)} - 2\mathbb{Q}_{xy}^{(1)} \\
G_{zzx} &= 2\mathbb{G}_x^{(2)} - 2\mathbb{Q}_{yz}^{(1)} \\
G_{zzy} &= 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)} \\
G_{yzx} &= 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\
\frac{G_{yyz}}{2} &= \mathbb{G}_z^{(2)} - \mathbb{Q}_{xy}^{(1)} \\
\frac{G_{zz y}}{2} &= \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)} \\
\frac{G_{xxz}}{2} &= \mathbb{G}_z^{(2)} + \mathbb{Q}_{xy}^{(1)} \\
G_{zxy} &= -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)} \\
\frac{G_{zzx}}{2} &= \mathbb{G}_x^{(2)} - \mathbb{Q}_{yz}^{(1)} \\
\frac{G_{xxy}}{2} &= \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)} \\
\frac{G_{yyx}}{2} &= \mathbb{G}_x^{(2)} + \mathbb{Q}_{yz}^{(1)}
\end{aligned}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & G_{yzz} \\ G_{zxx} & G_{zxy} & G_{zxx} \\ G_{xyx} & G_{xyy} & G_{xyz} \end{bmatrix}$$

$$G_{yzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$

$$G_{yzy} = \mathbb{G}_z^{(3)} + \mathbb{Q}_{xy}^{(2)}$$

$$G_{yzz} = -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$

$$G_{zxx} = -\mathbb{G}_z^{(3)} + \mathbb{Q}_{xy}^{(2)}$$

$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$

$$G_{zzz} = \mathbb{G}_x^{(3)} + \mathbb{Q}_{yz}^{(2)}$$

$$G_{xyx} = \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$

$$G_{xyy} = -\mathbb{G}_x^{(3)} + \mathbb{Q}_{yz}^{(2)}$$

$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} \\ G_{xxyy} & G_{yyyy} & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} \\ G_{xxzz} & G_{yyzz} & G_{zzzz} & G_{zzyz} & G_{zzzx} & G_{zzxy} \\ G_{xxyz} & G_{yyyz} & G_{zzyz} & G_{zyyz} & G_{yzzx} & G_{yzxy} \\ G_{xxzx} & G_{yyzx} & G_{zzzx} & G_{yzzx} & G_{zzzx} & G_{zzxy} \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & G_{yzxy} & G_{zzxy} & G_{xyxy} \end{bmatrix}$$

$$G_{xxxx} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)}$$

$$G_{xxyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)}$$

$$G_{xxzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + \mathbb{G}_v^{(1)}$$

$$G_{xxyz} = 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)}$$

$$G_{xxzx} = -\mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_y^{\beta(1)}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yyyy} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)}$$

$$G_{yyzz} = \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - \mathbb{G}_v^{(1)}$$

$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$

$$G_{yyzx} = \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_y^{\beta(1)}$$

$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzzz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)}$$

$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$

$$G_{zzzx} = \mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_y^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{yzyz} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - \mathbb{G}_v^{(2)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{yzxy} = \mathbb{G}_{xz}^{(2)} + 2\mathbb{G}_y^{\beta(1)}$$

$$G_{zxzx} = \mathbb{G}_0^{(2)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + \mathbb{G}_v^{(2)}$$

$$G_{zxxxy} = 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(2)}$$

$$G_{xyxy} = \mathbb{G}_0^{(2)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} \\ -G_{xxyy} & 0 & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} \\ -G_{xxzz} & -G_{yyzz} & 0 & G_{zzyz} & G_{zzzx} & G_{zzxy} \\ -G_{xxyz} & -G_{yyyz} & -G_{zzyz} & 0 & G_{yzzx} & G_{yzxy} \\ -G_{xxzx} & -G_{yyzx} & -G_{zzzx} & -G_{yzzx} & 0 & G_{zxxxy} \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & -G_{yzxy} & -G_{zxxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxyy} &= -2\mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xy}^{(1)} \\
G_{xxzz} &= 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(3)} - 4\mathbb{Q}_{xyz}^{(1)} \\
G_{xxyz} &= \mathbb{G}_{yz}^{(3)} - 4\mathbb{Q}_x^{\beta(1)} \\
G_{xxzx} &= \mathbb{G}_{xz}^{(3)} - 2\mathbb{Q}_y^{(1)} + 2\mathbb{Q}_y^{\alpha(1)} + 2\mathbb{Q}_y^{\beta(1)} \\
G_{xxxy} &= \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)} \\
G_{yyzz} &= 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(3)} + 4\mathbb{Q}_{xyz}^{(1)} \\
G_{yyyz} &= \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_x^{(1)} - 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)} \\
G_{yyzx} &= \mathbb{G}_{xz}^{(3)} - 4\mathbb{Q}_y^{\beta(1)} \\
G_{yyxy} &= \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)} \\
G_{zzyz} &= \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)} \\
G_{zzzx} &= \mathbb{G}_{xz}^{(3)} + 2\mathbb{Q}_y^{(1)} - 2\mathbb{Q}_y^{\alpha(1)} + 2\mathbb{Q}_y^{\beta(1)} \\
G_{zzxy} &= \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)} \\
G_{yzzx} &= -\mathbb{Q}_z^{(1)} - 4\mathbb{Q}_z^{\alpha(1)} \\
G_{yzxy} &= \mathbb{Q}_y^{(1)} + 4\mathbb{Q}_y^{\alpha(1)} \\
G_{zxxxy} &= -\mathbb{Q}_x^{(1)} - 4\mathbb{Q}_x^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{yzzx} & G_{zxzx} & G_{zxxy} \\ G_{yzxy} & G_{zxxxy} & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{yzyz} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(4)} \\
G_{yzzx} &= 2\mathbb{G}_{xy}^{(4)} \\
G_{yzxy} &= 2\mathbb{G}_{xz}^{(4)} \\
G_{zxzx} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 2\mathbb{G}_v^{(4)} \\
G_{zxxxy} &= 2\mathbb{G}_{yz}^{(4)} \\
G_{xyxy} &= \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & G_{yzxy} \\ -G_{yzzx} & 0 & G_{zxxy} \\ -G_{yzxy} & -G_{zxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{yzzx} &= -\mathbb{Q}_z^{(2)} \\
G_{yzxy} &= \mathbb{Q}_y^{(2)} \\
G_{zxxxy} &= -\mathbb{Q}_x^{(2)}
\end{aligned}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} G_{xxyz} & G_{xxzx} & G_{xxxy} \\ G_{yyyz} & G_{yyzx} & G_{yyxy} \\ G_{zzyz} & G_{zzzx} & G_{zzxy} \\ G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{zxxy} & G_{zxzx} & G_{zxxy} \\ G_{xyyz} & G_{xyzx} & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{xxyz} &= \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{(4)} + 2\mathbb{Q}_x^{\alpha(2)} \\
G_{xxzx} &= 2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(3)} - \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)} \\
G_{xxxy} &= -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)} \\
G_{yyyz} &= -2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)} \\
G_{yyzx} &= \mathbb{Q}_y^{(3)} + 2\mathbb{Q}_y^{(4)} + 2\mathbb{Q}_y^{\alpha(2)} \\
G_{yyxy} &= 2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)} \\
G_{zzyz} &= 2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)} \\
G_{zzzx} &= -2\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(3)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)} \\
G_{zzxy} &= \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)} \\
G_{yzyz} &= -3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xy}^{(2)} \\
G_{yzzx} &= -\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)} \\
G_{yzyx} &= \mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(4)} - \mathbb{Q}_y^{\alpha(2)} + \mathbb{Q}_y^{\beta(2)} \\
G_{zxyz} &= \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)} \\
G_{zxzx} &= 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(5)} + \mathbb{Q}_{xy}^{(2)} \\
G_{zxyx} &= -\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)} \\
G_{xyyz} &= -\mathbb{G}_{xz}^{(5)} + \mathbb{Q}_y^{(4)} - \mathbb{Q}_y^{\alpha(2)} - \mathbb{Q}_y^{\beta(2)} \\
G_{xyzx} &= \mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)} \\
G_{xyxy} &= 2\mathbb{G}_v^{(5)} + \mathbb{Q}_{xy}^{(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzzx} & G_{yzxy} \\ G_{zxzx} & G_{zxyy} & G_{zxzz} & G_{zxyz} & G_{zxzx} & G_{zxxy} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & G_{xyyz} & G_{xyzx} & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
G_{yzxx} &= \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)} \\
G_{yzyy} &= -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{yzzz} &= 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{yzyz} &= -3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xy}^{(3)} \\
G_{yzzx} &= \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxy} &= -\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxzx} &= 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxyy} &= \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)} \\
G_{zxxx} &= -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{zxyz} &= -\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxzx} &= 3\mathbb{G}_u^{(6)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xy}^{(3)} \\
G_{zxxy} &= \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzx} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)} \\
G_{xyyz} &= \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyzx} &= -\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyxy} &= 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xy}^{(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & G_{xxyz} & G_{xxzx} & G_{xxxy} & G_{xxzy} & G_{xxxz} & G_{xxyx} \\ G_{yyxx} & G_{yyyy} & G_{yyzz} & G_{yyyz} & G_{yyzx} & G_{yyxy} & G_{yyzy} & G_{yyxz} & G_{yyyx} \\ G_{zzxx} & G_{zzyy} & G_{zzzz} & G_{zzyz} & G_{zzzx} & G_{zzxy} & G_{zzzy} & G_{zzxz} & G_{zzyx} \\ G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzzx} & G_{yzxy} & G_{yzzy} & G_{yzxz} & G_{yzyx} \\ G_{zxxx} & G_{zxyy} & G_{zxzz} & G_{zxyz} & G_{zxxz} & G_{zxxxy} & G_{zxzy} & G_{zxxx} & G_{zxyx} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & G_{xyyz} & G_{xyzx} & G_{xyxy} & G_{xyzy} & G_{xyxz} & G_{xyyx} \end{bmatrix}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & G_{yzyz} & G_{yzxz} & G_{yzxy} & G_{yzzy} & G_{yzxz} & G_{yzyx} \\ G_{zxxx} & G_{zxyy} & G_{zxzz} & G_{zxzy} & G_{zxzx} & G_{zxyx} & G_{zxzy} & G_{zxzx} & G_{zxyx} \\ G_{xyxx} & G_{xyyy} & G_{xyyz} & G_{xyyz} & G_{xyzx} & G_{xyxy} & G_{xyzy} & G_{xyxz} & G_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
G_{yzzx} &= \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{(3)} \\
G_{yzyy} &= -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{yzzz} &= 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{yzyz} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzxz} &= 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxy} &= 2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{yzzy} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} - 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{yzxz} &= -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzyx} &= -2\mathbb{G}_{xz}^{(4)} - \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxzx} &= 2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} - \mathbb{Q}_y^{\beta(3)} \\
G_{zxyy} &= \mathbb{Q}_y^{(5)} + 2\mathbb{Q}_y^{(6)} + 2\mathbb{Q}_y^{\alpha(3)} \\
G_{zxzz} &= -2\mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(5)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{zxyz} &= 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxzx} &= \mathbb{G}_0^{(3)} - 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} - 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxxy} &= 2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{zxzy} &= -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxzz} &= -\mathbb{G}_0^{(3)} + 2\mathbb{G}_u^{(4)} + 3\mathbb{G}_u^{(6)} + 2\mathbb{G}_v^{(4)} - \mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{zxyx} &= -2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzz} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)} \\
G_{xyyz} &= 2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} - \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyzx} &= 2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyxy} &= \mathbb{G}_0^{(3)} + 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)} \\
G_{xyzy} &= -2\mathbb{G}_{xz}^{(4)} + \mathbb{G}_{xz}^{(6)} + \mathbb{Q}_y^{(2)} + \mathbb{Q}_y^{(6)} - \mathbb{Q}_y^{\alpha(3)} + \mathbb{Q}_y^{\beta(3)} \\
G_{xyxz} &= -2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyyx} &= -\mathbb{G}_0^{(3)} - 4\mathbb{G}_u^{(4)} + 2\mathbb{G}_v^{(6)} + \mathbb{Q}_{xyz}^{(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\left[\begin{array}{ccc} G_{xxxx} & G_{xxyy} & G_{xxzz} \\ G_{yyyy} & G_{yyyy} & G_{yyyz} \\ G_{zzzz} & G_{zzzy} & G_{zzzz} \\ G_{yyzz} & G_{yyzy} & G_{yyzz} \\ G_{zzxx} & G_{zzxy} & G_{zzxz} \\ G_{xxyx} & G_{xxyy} & G_{xxyz} \\ G_{yyzz} & G_{yyzy} & G_{yyzz} \\ G_{zzxx} & G_{zzxy} & G_{zzxz} \\ G_{xxyy} + G_{yyzz} - G_{yyzy} - G_{zzxz} + G_{zzxx} & G_{xyyy} & G_{xyyz} \\ -\frac{G_{yyyy}}{2} + \frac{G_{yyzz}}{2} + G_{xxyy} & G_{xyyz} + \frac{G_{zzzz}}{2} - \frac{G_{zzxz}}{2} & \frac{G_{xxxx}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzzy} \end{array} \right]$$

$$\begin{aligned}
G_{xxxx} &= 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} + 2\mathbb{G}_v^{(1)} + 4\mathbb{G}_v^{(2)} \\
G_{xxxxy} &= \mathbb{G}_{4z}^{(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{xxxz} &= -\mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} - \mathbb{G}_y^{\beta(1)} \\
G_{yyyyx} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{yyyxy} &= 3\mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} + 2\mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 4\mathbb{G}_u^{(2)} - 2\mathbb{G}_v^{(1)} - 4\mathbb{G}_v^{(2)} \\
G_{yyyz} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{zzzx} &= \mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} - 2\mathbb{G}_{xz}^{(5)} - \mathbb{G}_y^{\beta(1)} \\
G_{zzzy} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{zzzz} &= 3\mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} + 2\mathbb{G}_4^{(1)} + 4\mathbb{G}_u^{(1)} + 8\mathbb{G}_u^{(2)} \\
G_{yyzx} &= \mathbb{G}_{xz}^{(1)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_y^{\beta(1)} \\
G_{yyzy} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} \\
G_{yyzz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} + 3\mathbb{G}_u^{(3)} - \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} \\
G_{zzxx} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(1)} - 3\mathbb{G}_u^{(3)} + \mathbb{G}_v^{(1)} + \mathbb{G}_v^{(3)} \\
G_{zzxy} &= \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} \\
G_{zzxz} &= \mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} + \mathbb{G}_{xz}^{(3)} + 2\mathbb{G}_{xz}^{(5)} - \mathbb{G}_y^{\beta(1)} \\
G_{xxyx} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{xxyy} &= \mathbb{G}_0^{(1)} + 2\mathbb{G}_{4u}^{(1)} - \mathbb{G}_4^{(1)} - 2\mathbb{G}_u^{(1)} - 2\mathbb{G}_v^{(3)} \\
G_{xxyz} &= 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} \\
G_{yzzx} &= \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} \\
G_{yzzy} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} + \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} + 3\mathbb{G}_u^{(5)} - \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} \\
G_{yzzz} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} \\
G_{zxxx} &= -\mathbb{G}_{4y}^{\alpha(1)} + \mathbb{G}_{xz}^{(1)} + 2\mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(3)} - \mathbb{G}_y^{\beta(1)} \\
G_{zxyx} &= 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} \\
G_{zxxz} &= \mathbb{G}_0^{(1)} - \mathbb{G}_{4u}^{(1)} - \mathbb{G}_{4v}^{(1)} - \mathbb{G}_4^{(1)} + \mathbb{G}_u^{(2)} - 3\mathbb{G}_u^{(5)} + \mathbb{G}_v^{(2)} + \mathbb{G}_v^{(5)} \\
G_{xyyy} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)} \\
G_{xyyz} &= \mathbb{G}_{xz}^{(2)} - \mathbb{G}_{xz}^{(5)} + 2\mathbb{G}_y^{\beta(1)}
\end{aligned}$$